HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged **Linux kernel modules** and low-level components for endpoint security, supporting **Red Hat, Debian, and Ubuntu** distributions.
- Implemented and maintained **C/C++** code for critical cybersecurity features, focusing on system calls, memory management, and process monitoring.
- Designed and validated **user-space APIs** that interact with kernel-level functionality, ensuring seamless integration and data flow.
- Debugged complex kernel, performance, and system-level issues across **x86_64 and ARM64** architectures, identifying root causes for stability.
- Contributed to **CI/CD pipelines** for automated kernel testing and build processes, enhancing deployment efficiency and reliability.
- Collaborated closely with **QA, DevOps, and security teams** to ensure product integrity and rapid resolution of identified vulnerabilities.
- Supported platform enablement and distribution porting efforts, extending the reach of the security product to new Linux environments.

SKILLS

Languages – C, C++, Python, Java, SQL, R

Kernel Development – Linux Kernel Modules, eBPF, System Calls, Filesystems, Networking, Memory Management, Process Monitoring, Concurrency

Operating Systems – Linux (Red Hat, Debian, Ubuntu, SUSE), x86_64, ARM64

Development Tools – Git, Jenkins, Docker, C/C++ Compilers (GCC/Clang), GDB, Make

Cybersecurity Concepts – Endpoint Security, Application Control, Threat Prevention, EDR, Antivirus

Cloud & Virtualization – AWS (EC2, S3, Lambda, CloudWatch), KVM, VMware

Databases & Data Handling – PostgreSQL, MySQL, Oracle, pandas, NumPy, scikit-learn

PROJECTS

Linux Kernel Module for System Call Interception

- Developed a **Linux kernel module** in **C** to intercept and analyze specific system calls, enhancing security monitoring capabilities for enterprise endpoints.
- Implemented low-level logic for **process monitoring** and **memory management** within the kernel module, ensuring efficient resource utilization.
- Designed a corresponding user-space **Python utility** to configure and communicate with the kernel module, facilitating dynamic policy updates.

eBPF-based Network Traffic Analysis Tool

- Created an **eBPF** program to monitor and filter network traffic at the kernel level, identifying anomalous communication patterns for threat detection.
- Integrated with **x86_64 and ARM64** architectures, ensuring broad compatibility across server and embedded Linux systems.
- Developed a **C/C++** backend service to process eBPF data and expose insights via a REST API, supporting real-time security dashboards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)